

Server stress-test report

Script and what it does:

The script used to stress test the server is a lightweight, failsafe, and configurable script that monitors server utilization and writes csv and log output in created telemetry folders at root.

It collects:

- CPU usage (%) → via mpstat
- Load average (1 minute) → from /proc/loadavg
- Memory usage (MB used / total) → via free -m
- Uptime (seconds) → from /proc/uptime
- Optional service health (active/inactive) → via systemctl is-active
- Timestamped CSV output Logs for monitoring events

Server hardware:

- CPU: 2 core Intel CPUs, single threaded per core
- Memory: 3.8 GB RAM total, ~2.4 GB free at idle
- Disk: 78 GB total, 12% used

Testing:

Simulated 10 separate users by opening independent browser profiles with unique cookies and sessions with limitation being same ip origin. Each user collaborated in 5 total pairs of 2, with 4 pairs running Python, and 1 pair running HTML. All python groups ran moderate code, with 1 pair performing heavy computation and eventually an infinite loop to push the server.

Server experienced:

- 10 collaborative TLJH users
- CPU saturation
- RAM pressure
- Infinite loops
- No swap (zero virtual memory fallback)

Server monitor results:

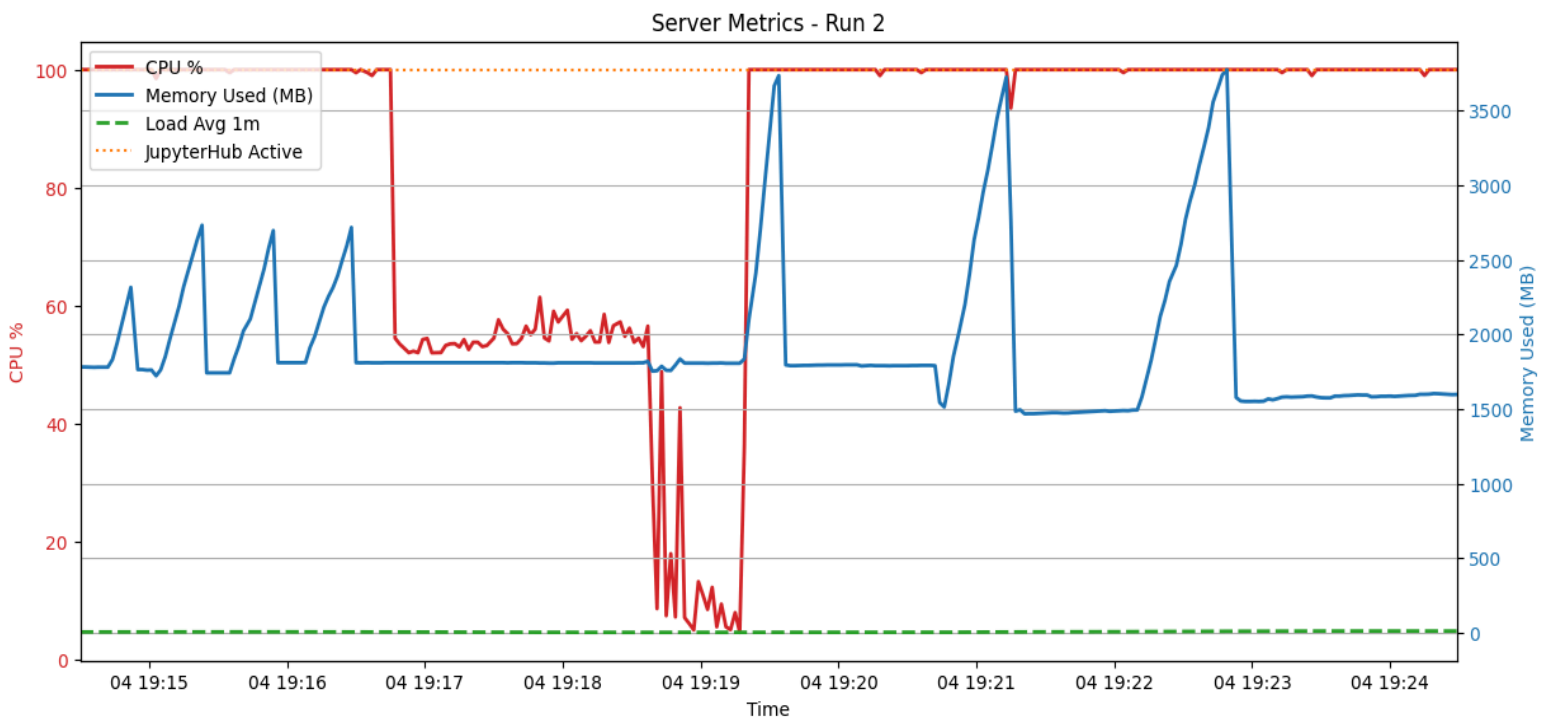
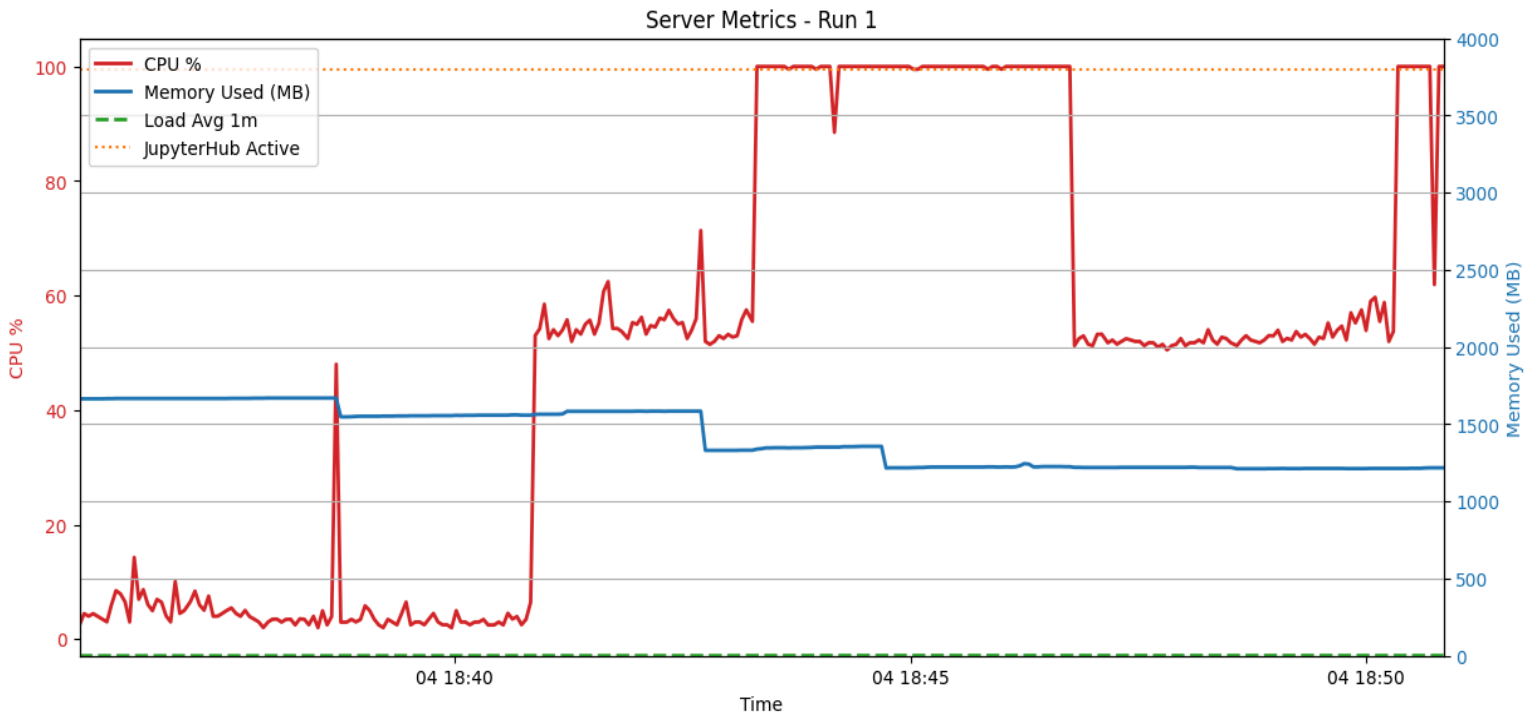
First run (15 minutes):

- Total samples: 298 at 2 sec interval
- Peak CPU: 100%
- Load Avg (1m): 2.23 / 2 cores
- RAM: 1.6 GB / 3.8 GB

Second run (10 minutes, heavy computation + infinite loops):

- Total samples: 294
- Max CPU: 100%
- Load Avg (1m): 10.56 / 2 cores suggesting 8 processes waiting for CPU at all times
- RAM: 3.7GB / 3.8 GB

Graphs:



TLJH kernel performance (not the server):

- 10 users collaborated concurrently in 5 rooms
- Average memory usage per kernel 250-400MB with moderate coding for python and 150-200MB for html editing
- Infinite loop effects: Memory spiked to 2.2 GB, then stabilized to ~300 MB. Real-time collaboration (RTC) drastically slowed but persisted

Note: only possible with admin privilege users, non-admins cannot access other user's collab rooms or invite links

Recommended server:

- CPU: 4-core. Average load per 2-core VM reached 100% in heavy scenarios. 4 cores would provide enough headroom for 30 users doing moderate work
- Memory: 16GB RAM. Observed peak memory usage per user: ~0.37 GB per heavy Python usage. Scaling to 30 students: $0.37 \times 30 \approx 11.1$ GB $\rightarrow$ 16 GB recommended for safe margin